Gil Jorge

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EXPERIENCE AND EDUCATION

Aqua Hotels, Sales/Marketing, Lagoa— Paid *Internship*

July 2022

Managed reservations, developed an automated excel workflow for retrieving the accounting data and visualizing it in dashboards. Client service through managing the website's chatbot and direct problem and process resolving.

Champalimaud Center for the Unknown, Lisbon— *Internship*

Jan 2026-present

6 month internship at Paton's Learning lab. Currently working on the project: **Clockless Alignment: Developing an Algorithm to Infer Neural–Behavior Correspondence.** where I temporally align eletrophysiological data through hardware artifacts and biological patterns with state machine data from the same experiment. Studying the intersection of natural/artificial intelligence.

Instituto Superior Técnico, Lisbon— *Bsc Physics Engineering*

Sep 2023 - July 2026 GPA out of 20: Year 1- 16; Year 2- 15; Year 3- 17

Completing **210 credits** (ECTS) instead of the usual 180 by taking extra courses (Abstract Algebra, Statistical Physics, Knowledge Representation and Reasoning of Msc CS, Cryptography and Security Protocols and Mathematics for Machine Learning of Msc Applied Math). Other courses: Deep learning of Msc CS 17/20 , Discoveries in Modern Physics, 19/20 for researching and presenting on Quantum Information and Computation, and a project at the Quantum Photonics Lab with 19/20. **One of 7 students** of my cohort to earn a spot at oral exam in calculus (went in multivariable calculus, calculus 2).

SKILLS (software + languages)

Python (numpy, matplotlib, pytorch), **C++** (root), **R**, **Latex**
Emacs Org mode
Linux- Lubuntu in a 2012 laptop, maximizing it's efficiency software wise for it's hardware, now became a self hosted server on Nextcloud. Using Mint on new laptop.
Adobe Premiere; Davinci Resolve
Portuguese; C1 English; Advanced Spanish; Intermediate French.

HONORS

-Featured on Portimão Jornal for **maximum grades** (20/20) in all courses in high school.
-Level 5 music formal education (theory+guitar+chorus)
-Discover EU ambassador
-Part of the class featured in national journals for the making of national gardens day.

PROJECTS

Writing (https://wayofthe0.wordpress.com/):
500 thousand written words, on personal website preparing selected pieces for publication on public platforms. Mostly philosophy.

Multisport athlete (https://www.youtube.com/@giljorge7479):
Compete in national track and field championships representing the student union, represented Sporting Clube de Portugal for 2 years. Also competing in Greco-Roman/freestyle wrestling at the national level, recently obtaining 4th place at national championships. Non competitively practicing calisthenics/streetlifting, olympic weightlifting, powerlifting. Highlights of the last ones include human flag, one arm pull up... and are available online along with video editing projects.

Digital brain (These and other coding projects available on github.com/giljorge0)**:**
Built a provenance-aware neuro-symbolic memory system with various memory layers for continual personal knowledge consolidation. Implements an ingestion pipeline (Org-mode, LLM chat history...); combines dense vector retrieval with a symbolic knowledge graph (NetworkX/SQLite, visualized as D3.js) with LLM-extracted relations; a continual consolidation loop running Louvain community detection, PageRank centrality, duplicate merging, contradiction flagging, and confidence decay; a LangGraph-powered query orchestrator routing across graph traversal, vector search, and temporal reasoning with human-in-the-loop fallback; a Gap Agent that finds knowledge gaps and suggests material; corpus character model for writing style analysis and idea completion in the user's own voice; and a benchmark comparing against flat RAG and GraphRAG baselines.

Emmergent AI-AI communication and geometry of meaning research programme:
Designed eight experiments on channel failure modes, zero-knowledge causal/information theoretic communication metrics, multilingual concept gaps via LaBSE centroid geometry, translation validity, interlingua centroids, UMAP-based universal language bootstrapping, and dense embedding-based communication with higher information density than text.